



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

June 02, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 8178
(SIXTEENTH REVISION)

EXPIRATION DATE: 2029-05-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: National Aeronautics and Space Administration
Washington, DC
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the transportation in commerce of certain non-DOT specification cylinders containing oxygen, compressed. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.302(a)(1) in that non-DOT specification cylinders are not authorized and 173.301(f) in that pressure relief devices are required, except as specified herein.
5. BASIS: This special permit is based on the application of the National Aeronautics and Space Administration dated January 27, 2025, submitted in accordance with § 107.105, and the public proceeding thereon.

Tracking Number: 2025015301

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Oxygen, compressed	2.2	UN1072	N/A

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Prescribed packaging is a non-DOT specification cylinder conforming with the following design:

Stainless steel cylinders consisting of two formed and machined hemispheres welded to a cylindrical section described as portable life support systems (PLSS). The cylinder must conform to specification SVHS 2840 on file with the Office of Hazardous Materials Safety (OHMS) and applicable referenced standards with a volumetric capacity of 240 cubic inches nominal, and a rated pressure of 1050 psig at 70 °F. The cylinder must also conform with United Technologies' drawing SV 784099. on file with OHMS. These cylinders need not be equipped with safety relief devices. Only 40 cylinders made to this design may be used under the terms of this special permit.

- b. OPERATIONAL CONTROLS: Only 40 stainless steel cylinders made in accordance with terms of this special permit (and application) may be used under the terms of this special permit. Prototype cylinders must have been tested and passed the qualification tests as noted in the application prior to first shipment.

8. SPECIAL PROVISIONS:

- a. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package and it is reoffered for transportation in conformance with this special permit and the HMR.

- b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - c. The steel cylinder described in paragraph 7.a. of this special permit may not be used after 43 years from the date of manufacture or 375 pressurizations, whichever comes first.
 - d. The cylinders must be used exclusively as a component of the international space station.
 - e. Each cylinder must be shipped in strong outside packagings in accordance with § 173.301(a)(9).
 - f. The transportation of oxygen by aircraft is only authorized when in accordance with § 175.501.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo aircraft only.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- "The Hazardous Materials Safety and Security Reauthorization Act of 2005" (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: TG/AS